The first critical symmetric factors

Human Compression project: a component of the L019 recovery

We retain the actual one-point attachment reduction. The remaining configuration has a fractured natural symmetric factor P of even degree m and rank $m/2+1$, and an even six-point rank-three A_5 factor E. Their supports meet only in the endpoint anchor a and their labels are consecutive disjoint intervals. This note settles $m \leq 8$. It makes no assertion about the remaining even degrees.

Lemma 1. *A fractured natural S_6 string C-marking of rank four with an endpoint anchor has an internal perfect split with one trivial internal factor and with both attaching groups odd.*

Proof. Reverse the marks and normalize the anchor to 1. Under its point stabilizer the first mark has one of the three representatives

$$(1\ 2), \quad (1\ 2)(3\ 4), \quad (1\ 2)(3\ 4)(5\ 6).$$

The earlier complete bounded computation tries every involution of $(S_6)_1$ at each later position. It requires distant commutation and adjacent noncommutation. The latter is necessary because the anchor forces the noncommutation diagram to be connected. It checks every pair of subset intersections, and requires the full group to have order 720. Thus it omits no anchored marking up to conjugacy; it need not identify all remaining equivalent tuples.

There are 32 terminal tuples in this enumeration, of which 24 have every maximal omission intransitive. For each such tuple and each internal position $j = 1, 2$, the verifier computes the actual omission orbits. It tries both choices of their order A, B and requires the prefix to be transitive on A and fix B pointwise, and the suffix to be transitive on B and fix A pointwise. It then requires a unique crossing $a_j b_j$ of the omitted mark. It tests pointwise that the remaining action on at least one side is trivial. When the A-action is trivial it tests oddness of the prefix and of the group generated by the crossing mark and suffix; when the B-action is trivial it uses the reversed pair of groups. Oddness is equivalent to at least one odd distinguished generator. All 24 fractured cases have a position passing these tests. The unchanged complete script, terminal receipt and output are retained from the earlier work; this proof does not extrapolate from a selected example. The tests are exactly the asserted conditions, proving the lemma. □

Lemma 2. *Let $(t_0, \ldots, t_4)$ be a fractured natural S_8 C-marking, with 1 moved only by t_0. If it occurs in an actual one-point attachment at 1 to a disjointly labelled transitive factor on at least two points, then t_0 is a pure transposition.*

Proof. Here the external factor need not be odd. Its original intersection property already gives the acquired even-kernel conditions

$$P_j \cap \mathrm{Alt}(X_{j-1}) \leq P_{j-1}, \qquad P_j = \langle t_0, \ldots, t_{j-1} \rangle, \quad X_j = 1^{P_j}. \tag{1}$$

We inspect the complete anchored degree-eight rank-five enumeration. Its search is the earlier full even-wreath enumerator with ambient group replaced by S_8 and target rank changed to five. It lists all nonidentity involutions literally, fixes point 1 after the first mark, and uses the exact point stabilizer and successive centralizers for conjugacy reduction. It tests all prefix subset intersections and the necessary order bound $[S_8 : H] \geq 2^{5-l}$ for a prefix of length l. Its completeness proof is unchanged: each conjugation fixes the previous marks and anchor, and

every independent next mark at least doubles the group order. At termination it requires order 40320. The node counts, including the empty prefix, are

$$1, 4, 60, 432, 90, 22.$$

Exactly nine terminal tuples generate S_8. The first six have a pure first mark. The ninth has a transitive maximal omission, so is excluded by fracture. The seventh and eighth have, with one-based table numbering, the following exact supported kernels:

Tuple	j	$\lvert P_j \cap \mathrm{Sym}(X_{j-1})\rvert$	$\lvert P_{j-1} \cap \mathrm{Sym}(X_{j-1})\rvert$	$\lvert P_j \cap \mathrm{Alt}(X_{j-1})\rvert$
7	3	120	10	60
8	4	720	60	360

Their previous supported kernels consist entirely of even permutations. Thus both violate (1), by index six even before their odd elements are considered. A separate literal-permutation implementation enumerates the groups, omission orbits and supported kernels of all nine tuples and verifies these orders and inclusions. This excludes the only two nonpure fractured candidates and proves the lemma. □

Theorem 3. *In the actual remaining P, E configuration described above, if $m \leq 8$, the whole marking has a valid internal rank and degree extension. For $m = 6$ it has an internal sufficient split inherited from P; for $m = 8$ the original junction split has valid leaves. The only critical symmetric degrees still unpaid are $m \geq 10$.*

Proof. For $m = 6$, Lemma 1 and the acquired one-point transport lemma give a sufficient internal perfect split in the whole marking. The acquired odd-attachment theorem then supplies its extension.

For $m = 8$, Lemma 2 applies with the actual external E. The pure first mark gives the entire leaf criterion for P. The leaf criterion for E follows from its odd external group P. The tail of E is transitive on the five points other than a, by the accepted small-even boundary theorem. Its first mark is $(a\ b)\beta$, with β supported among those five points. Hence the junction is the internal perfect split in the repaired two-sided extension theorem, with precisely these two valid leaves. That theorem increases degree and rank by one and gives the full symmetric group on the enlarged domain.

For completeness, $m = 2$ would require two independent marks in S_2 and is impossible. At $m = 4$, normalize the anchor to 1. The last two marks are transpositions on the other three points and must not commute. Write them $(b\ c), (c\ d)$ after relabelling. The first mark moves 1 and commutes with $(c\ d)$; it is either $(1\ b)$ or $(1\ b)(c\ d)$. In the second case, the first two marks act transitively on the four points, contradicting fracture of the last omission. In the first case all three marks form the simplex path and its middle mark is an internal pure perfect split. Transport and the acquired pure-split theorem finish this case. These are all positive even $m < 10$. □

Attribution and boundary. The principal source is Cameron–Fernandes–Leemans, *The number of string C-groups of high rank*, arXiv:2212.12723v2. We acquire the reviewed one-point transport, even-kernel and exact leaf criteria, and the six-point A_5 reduction; their original group-theoretic and software premises remain bound. The degree-six search is reused unchanged. An earlier degree-eight centralizer diagnostic traversed this same rank but retained only centralizer counterexamples; the present complete terminal list and literal leaf filter are retained separately. No conjecture, whole L019, Lean theorem or complexity measurement is claimed. The general critical leaf argument and the reversible stability correspondence are still required.